$P_{ave} = V_{rms} I_{rms} \cos(\phi)$ $\quad \phi = \theta_v - \theta_i$

$V_{rms} = \frac{|V|}{\sqrt{2}}$ $\quad I_{rms} = \frac{|I|}{\sqrt{2}}$

$S = \frac{1}{2}(V \cdot I^*) = (V_{rms} \cdot I^*_{rms})$

$P_L = \frac{1}{8} \frac{|V_s|^2}{R_s}$

Drawbacks of Nonlinearity

(1) Harmonic distortion

$y = a x_1 + a_2 x^2 + a_3 x^3$

$(\cos \omega t)^2 = \frac{1}{2}(1 + \cos 2\omega t)$

$(\cos \omega t)^3 = \frac{1}{4}(3 \cos \omega t + \cos 3\omega t)$

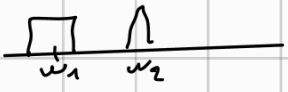
(2) Gain compression

$$\text{Gain} = \frac{V_{out}(\text{fund})}{V_{in}} = \frac{a_1 A + \frac{3 a_3 A^3}{4}}{A}$$

$\text{Gain} = a_1 + \frac{3}{4} a_3 A^2$

1) $a_1 > 0, a_3 < 0 \rightarrow \text{Gain} = a_1 - \frac{3}{4}|a_3| A^2 \rightarrow$ Gin

(3) Desensizitation



$x(t) = A_1 \cos \omega_1 t + A_2 \cos \omega_2 t$

$y = a_1 x + a_2 x^2 + a_3 x^3$

$\cos a \cdot \cos b = \frac{\cos(a+b) + \cos(a-b)}{2}$

$\frac{3}{4} a_3 A_1^3 \cos \omega_1 t + \frac{3 A_1 A_2^2}{2} a_3 \cos \omega_1 t$

$y_{BPF}(t) = \frac{3}{4} a_3 A_1^3 \cos \omega_1 t + \frac{3 A_1 A_2^2}{2} a_3 \cos \omega_1 t + a_1 A_1 \cos \omega_1 t$

if A_1 small

ignore $\frac{3}{4} a_3 A_1^3$

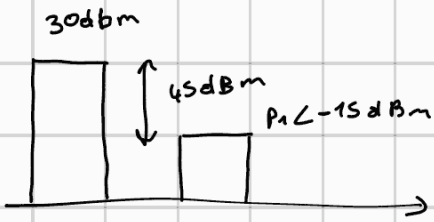
$\text{Gain} = \frac{(a_1 A_1 - \frac{3}{2}(a_3 A_1 A_2^2))}{A_1}$ $\quad \} A_2 \uparrow\uparrow$ No signal appears at output

Ex

$P_{Tx} = 1W = 30 dBm$

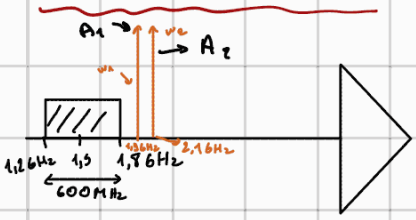
$P_1 - 10 < -25$

$P_1 < -15$



2. harmonik 45 dBm bastırılmalıdır

④ Intermodulation



$$A_{IM3} = \frac{3}{4} a_3 A_1^2 A_2 \left(\cos(2\omega_1 - \omega_2)t\right)$$

$$A_1 = A_2 = A \Rightarrow A_{IM3} = \frac{3}{4} a_3 A^3$$

system linear $\rightarrow a_1 A$

Intermodulation products $\rightarrow \frac{3}{4} a_3 A^3$

$$IM3 = \frac{\frac{3}{4} a_3 A^3}{a_1 A} = \frac{3}{4}\left|\frac{a_3}{a_1}\right| A^2$$

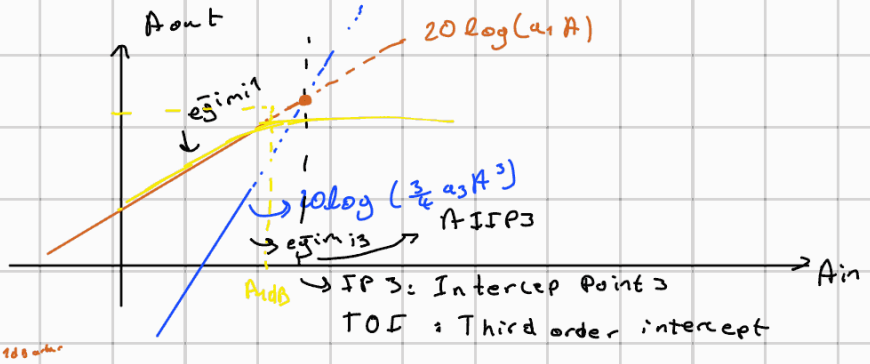
$$A_{IIP3} = 20\log\left(|a_1| A_{IIP3}\right) = 20\log\left(\frac{3}{4}|a_3| A_{IIP3}^3\right)$$

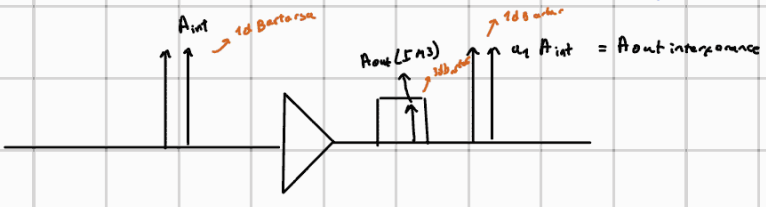
$$\boxed{A_{IIP3} = \sqrt{\frac{4}{3}\left|\frac{a_1}{a_3}\right|}}$$

$$A_{1dB} = \sqrt{0{,}145\left|\frac{a_1}{a_3}\right|}$$

$$A_{IIP3(dB)} - A_{1(dB)} = 9{,}6\,dB$$

A_{IIP3}↑ ⇒ more linear system



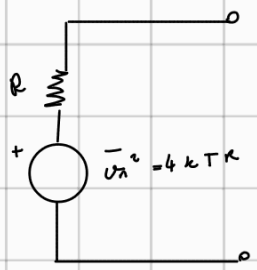


$$\frac{A_{out,int}(\omega_1,\omega_2)}{A_{out}(IM3)} = \frac{a_1 A_{int}}{\frac{3}{4} a_3 A_{int}^3} = \frac{a_1}{\frac{3}{4} a_3 A_{int}^2} = \frac{A_{IIP3}^2}{A_{int}^2}$$

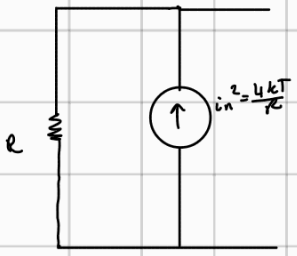
$$\boxed{A_{IIP3}^2 = \frac{A_{out,int}(\omega_1,\omega_2)}{A_{out}(IM3)} A_{int}^2}$$

Noise

Resistor Noise:



veya

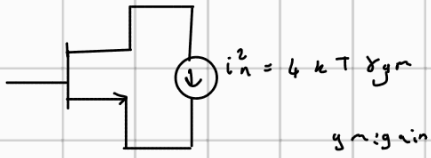


$1,3 \times 10^{-23}$ J/K

k: Boltzman sabiti

T = (kelvin) sıcaklık

MOS Transistor Noise



g_m: gain

γ: noise coefficient

